

SMD 0603, Industrial Grade NTC Thermistors



LINKS TO ADDITIONAL RESOURCES



QUICK REFERENCE DATA		
PARAMETER	VALUE	UNIT
Resistance value at 25 °C	10K to 100K	Ω
Tolerance on R_{25} -value	± 1	%
$B_{25/85}$ -value	3435 to 4311	K
Tolerance on $B_{25/85}$ -value	± 3	%
Maximum power dissipation at 25 °C P_{max25}	125	mW
Thermal time constant τ	≈ 5	s
Dissipation factor D	3	mW/K
Operating temperature range at zero power ⁽¹⁾	-40 to +150	°C
Storage temperature range	-40 to +150	°C
Weight	6	mg

Note

⁽¹⁾ Zero power is considered as measuring power maximum 1 % of P_{max25}

AGENCY APPROVALS

Agency approval documents, please see:

www.vishay.com/ppg?29251&documents

DESIGN-IN SUPPORT

For complete curve computation, please visit:

www.vishay.com/thermistors/ntc-rt-calculator/

FEATURES

- TCR ranging from -6.5 %/K at -40 °C to -2 %/K at 150 °C
- Tolerance on R_{25} of ± 1 %
- Suitable for wave or reflow soldering
- NiSn terminations
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



RoHS
COMPLIANT
HALOGEN
FREE

APPLICATIONS

- Temperature sensing, protection and compensation in industrial, telecom and consumer applications.

Examples are:

- Battery chargers
- Power supplies
- Office equipment
- LED compensation

This series is recommended for industrial applications, in particular the ones requiring high operation temperatures, and demanding high level of reliability for the products (similar to automotive standard AEC-Q200).

DESCRIPTION

Size 0603 (M1608) SMD chip thermistor with negative temperature coefficient (TCR) and matte tin (Sn) plated terminations. The device has no marking.

PACKAGING

Available in 8 mm punched paper tape on reel package of 4000 units.

CAUTIONS AND WARNINGS ON MOUNTING AND HANDLING

Please read the special instructions:

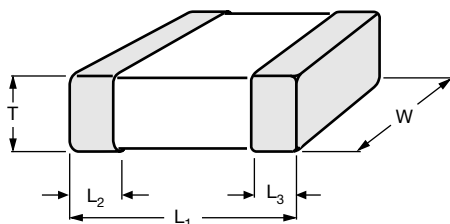
see www.vishay.com/doc?29224.

ELECTRICAL DATA AND ORDERING INFORMATION				
R_{25} (Ω)	R_{25} -TOL. (± %)	$B_{25/85}$ (K)	$B_{25/85}$ -TOL. (± %)	SAP MATERIAL AND ORDERING NUMBER
10 000	1	3950	3	NTCSI0603E3103*HHT ⁽¹⁾
10 000	1	3435	3	NTCSI0603E3103*LHT ⁽¹⁾
47 000	1	4108	3	NTCSI0603E3473*XHT ⁽¹⁾
100 000	1	4311	3	NTCSI0603E3104*XHT ⁽¹⁾

Note

⁽¹⁾ Replace digit * on R_{25} by F for tolerance ± 1 % on R_{25}

DIMENSIONS in millimeters

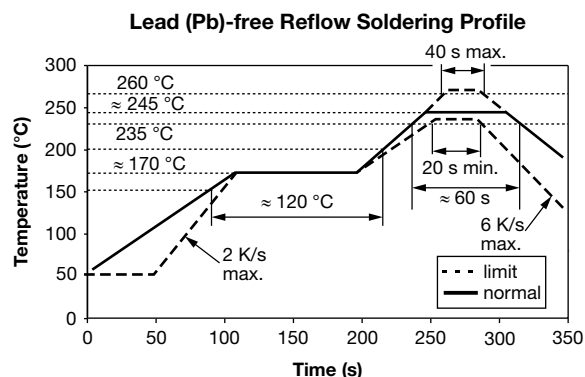
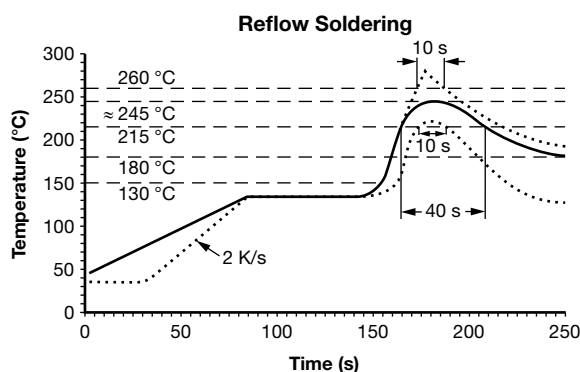


L ₁	W	T	L ₂ AND L ₃
1.6 ± 0.15	0.8 ± 0.15	0.8 ± 0.15	0.4 ± 0.2

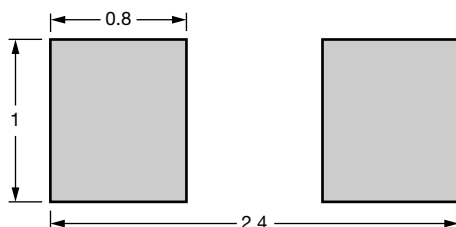
SOLDERING CONDITIONS

Soldering, handling, and mounting conditions are detailed in the instructions document: see www.vishay.com/doc?29224.

Typical examples of soldering processes that will provide reliable joints without damage, are shown below.



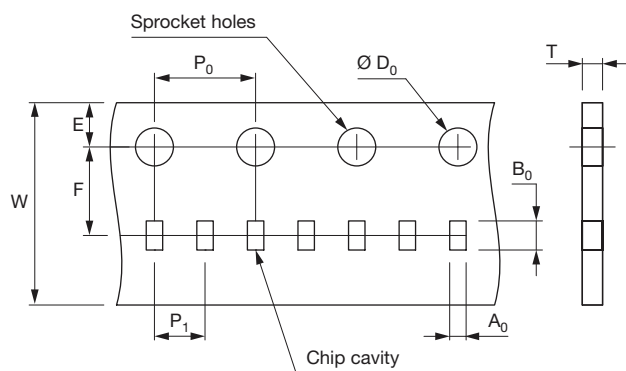
Recommended solder land pattern dimensions (mm)



PACKAGING

TAPE SPECIFICATIONS

All tape specifications are in accordance with IEC 60286-3. Basic dimensions are given below. Carrier tape material is paper.



DIMENSIONS OF PAPER TAPE in millimeters

PARAMETER	DIMENSION
A ₀ ⁽¹⁾	1.0 ± 0.2
B ₀ ⁽¹⁾	1.8 ± 0.2
W	8.0 ± 0.3
E	1.75 ± 0.1
F	3.5 ± 0.05
D ₀	1.55 ± 0.05
P ₀ ⁽²⁾	4.0 ± 0.1
P ₁	4.0 ± 0.1
T tape thickness max.	1.1

Notes

⁽¹⁾ Measured 0.3 mm above base pocket

⁽²⁾ P₀ pitch cumulative error over any 10 pitches ± 0.2 mm



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